MNIST Image Recognition

A compact convolutional neural network

Training, augmentation, and error analysis

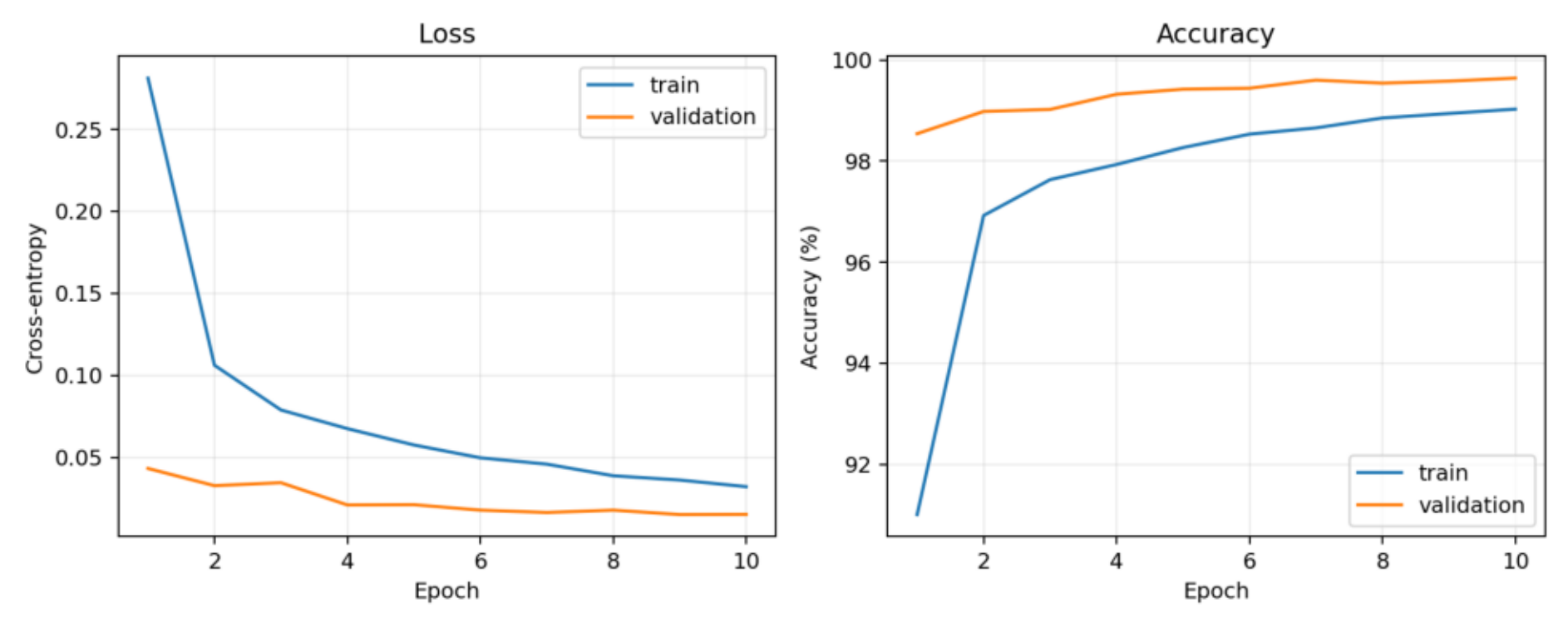
Experimental design

Strict separation between model development and final testing

1. Data: 60,000 MNIST training images and 10,000 test images
2. Training split: 55,000 train + 5,000 validation
3. Architecture: four convolution layers, batch normalization, dropout
4. Optimization: AdamW, cosine learning-rate schedule, cross-entropy loss
5. Bonus experiment: random affine and perspective augmentation on train only

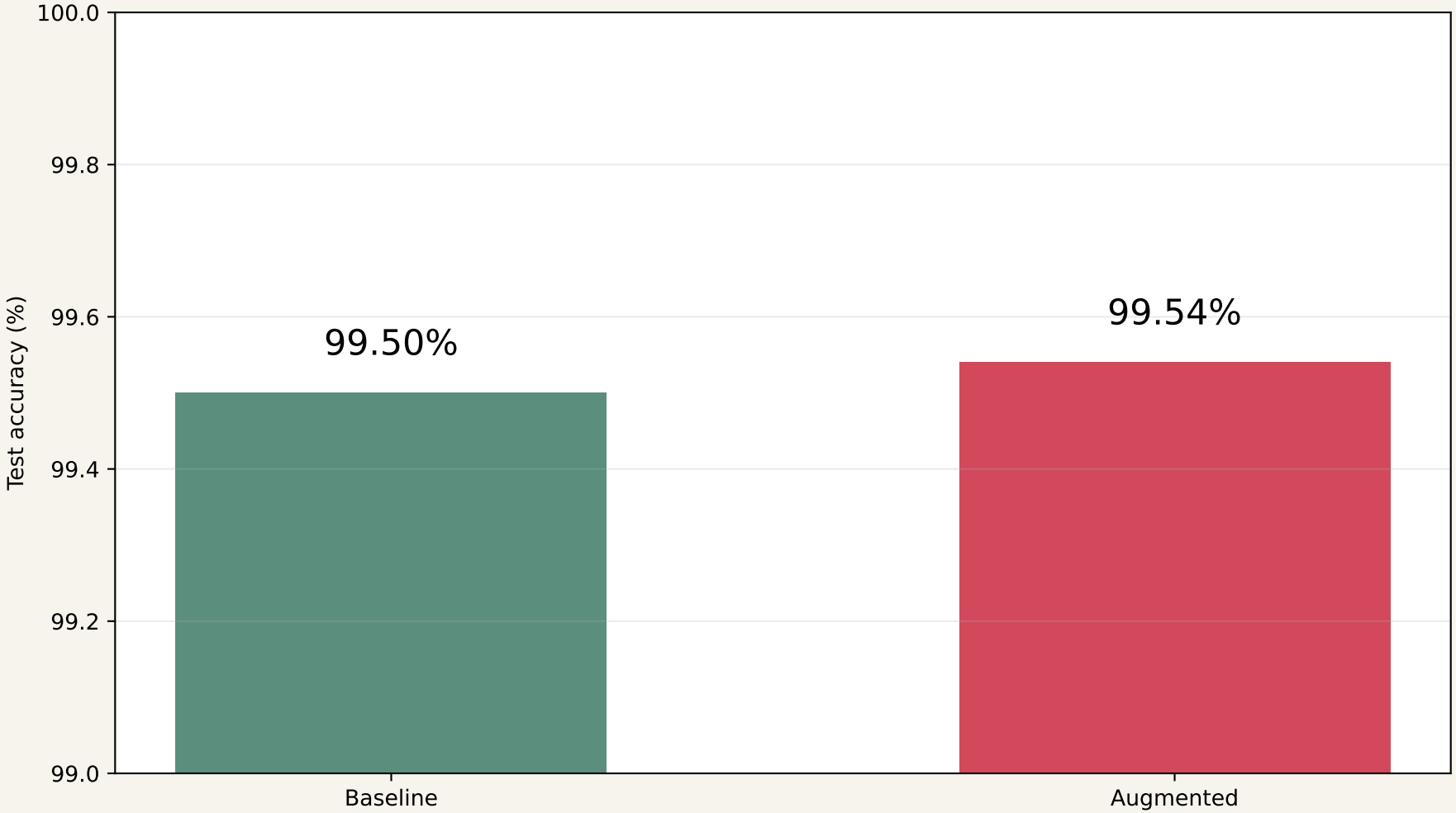
Learning behavior

Training and validation metrics for the augmented CNN



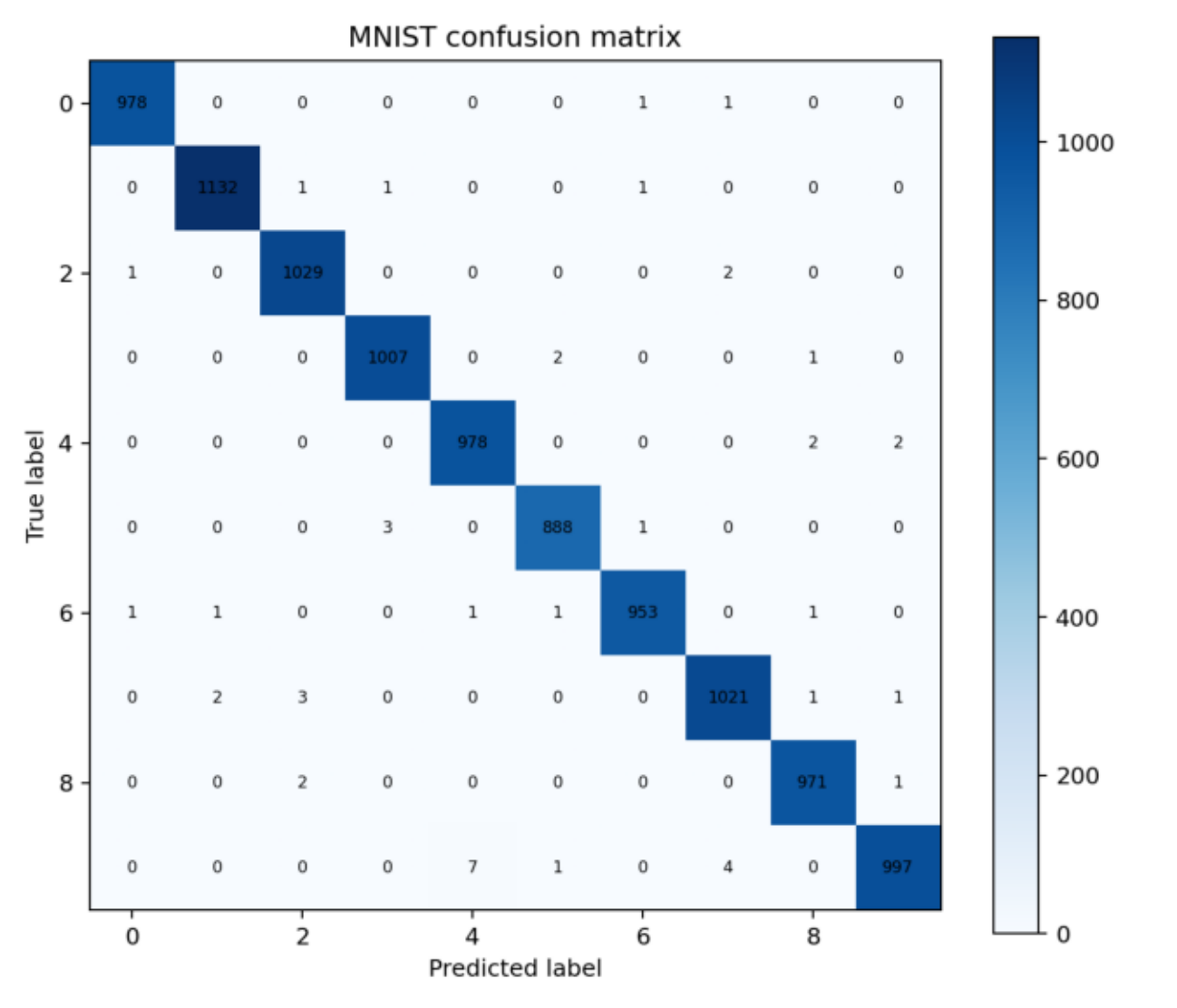
Test-set comparison

One final evaluation on the untouched official test split



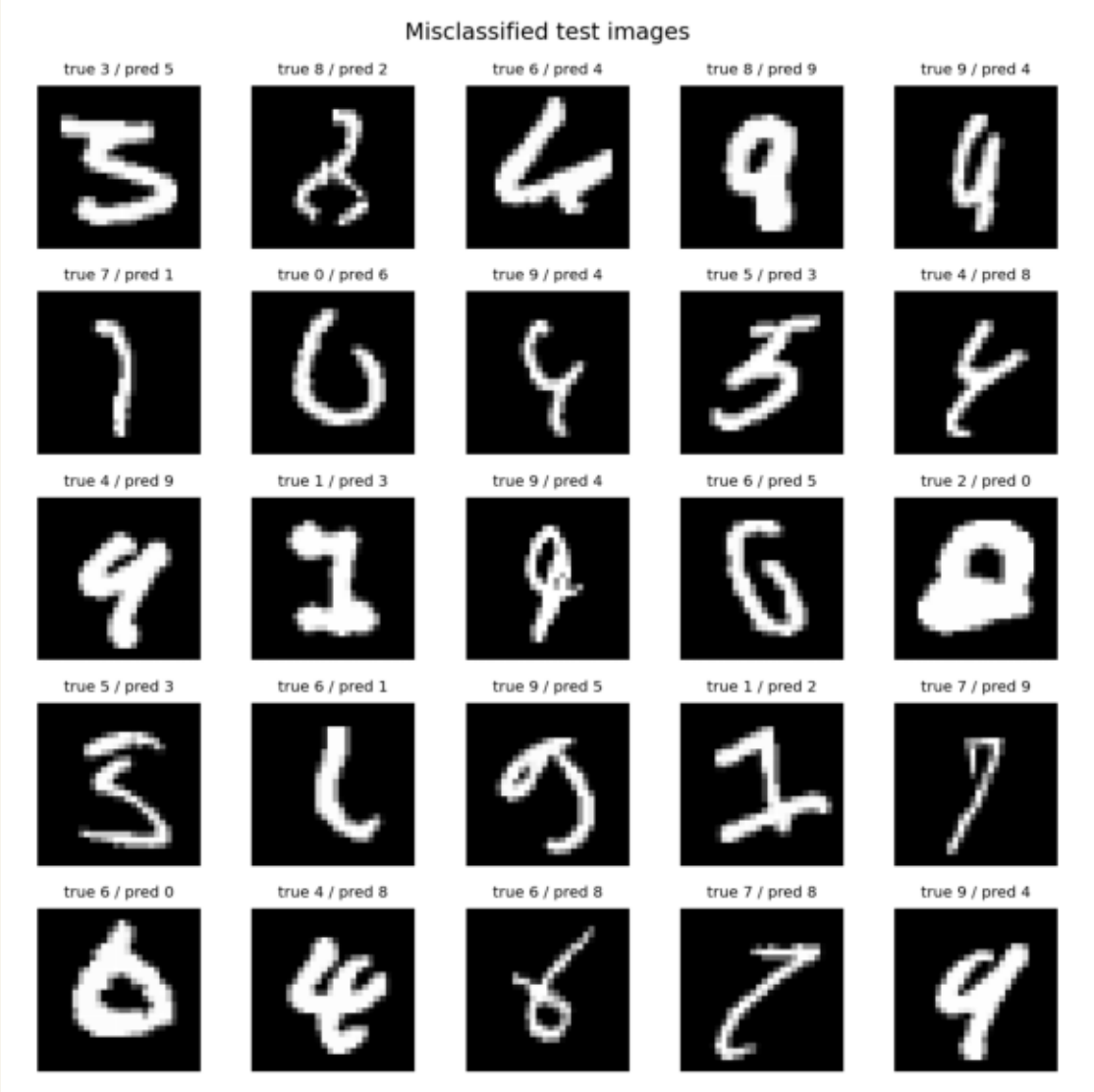
Where the model is confused

Rows are true classes; columns are predictions



Error analysis

Ambiguous handwriting accounts for many remaining mistakes



Conclusions

- The augmented CNN reached 99.54% test accuracy.
- Augmentation changed accuracy by +0.04 percentage points.
- The test split was never used for training or checkpoint selection.
- Saved JSON metrics and figures make the result reproducible and auditable.